

Sound #2

$$f = f_s \left(\frac{v \pm v_o}{v \mp v_s} \right)$$

speed of sound in room-temperature air = 340 m/s

1. A student is riding a bicycle at 10 m/s toward a stationary factory whistle that is blowing at a frequency of 800 Hz. What frequency does the student perceive?
2. A police car with a 600 Hz siren is chasing a speeding car. The police car moves at 40 m/s, while the speeding car moves away from the police at 35 m/s. What frequency does the driver of the speeding car hear?
3. *Challenge:* A bat uses echolocation to orient itself to its surroundings. One such bat flies at 12 m/s directly toward a large, stationary wall while emitting an ultrasonic pulse at a frequency of 30 000 Hz.
 - (a) What is the frequency of the sound waves as they reach the stationary wall?
 - (b) What frequency does the bat hear when those sound waves reflect off the wall and return to the bat?